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Solid-state imaging element

Abstract

A solid-state imaging element includes a first semiconductor substrate having a first semiconductor circuit on a first surface of the substrate, a second semiconductor substrate having a second semiconductor circuit on a second surface of the substrate, and a pixel substrate having a pixel circuit on one surface of the substrate, in which the first semiconductor substrate, the second semiconductor substrate, and the pixel substrate are joined to each other such that the first surface of the first semiconductor substrate and the second surface of the second semiconductor substrate face the one surface of the pixel substrate, and the first semiconductor circuit and the second semiconductor circuit are connected to each other on the first surface side and the second surface side, opposite to the side facing the pixel substrate.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10049915	12/2017	Fong	N/A	H01L 25/50
2015/0035109	12/2014	Kataoka	257/443	H01L 24/06
2017/0005128	12/2016	Sasaki	N/A	N/A
2017/0317061	12/2016	Takahashi et al.	N/A	N/A
2018/0114807	12/2017	Yamagishi et al.	N/A	N/A
2018/0138225	12/2017	Kim	N/A	H01L 27/14645
2018/0240836	12/2017	Sasaki	N/A	N/A
2020/0177829	12/2019	Takahashi	N/A	H04N 25/70

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105814670	12/2015	CN	N/A
107004672	12/2016	CN	N/A
2010-245506	12/2009	JP	N/A
10-2016-0098221	12/2015	KR	N/A
10-2017-0096102	12/2016	KR	N/A
201526193	12/2014	TW	N/A
201633502	12/2015	TW	N/A
201637190	12/2015	TW	N/A
201909440	12/2018	TW	N/A
2015/093313	12/2014	WO	N/A
2016/098691	12/2015	WO	N/A
2016/152577	12/2015	WO	N/A
2019/021705	12/2018	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion of PCT Application No. PCT/JP2020/009837, issued on Jun. 9, 2020, 09 pages of ISRWO. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Phase of International Patent Application No. PCT/JP2020/009837 filed on Mar. 6, 2020, which claims priority benefit of Japanese Patent Application No. JP 2019-045759 filed in the Japan Patent Office on Mar. 13, 2019. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

FIELD

(2) The present disclosure relates to a solid-state imaging element.

BACKGROUND

(3) There is a three-dimensional mounting technique of stacking a plurality of semiconductor substrates. For example, there is a known configuration of a solid-state imaging element in which a pixel substrate on which a pixel circuit is formed and a semiconductor substrate on which a semiconductor circuit such as a logic circuit is formed are stacked (refer to Patent Literature 1).

CITATION LIST

Patent Literature

(4) Patent Literature 1: JP 2010-245506 A

SUMMARY

Technical Problem

(5) When stacking a plurality of semiconductor substrates on a pixel substrate in the structure disclosed in Patent Literature 1, there is a concern of occurrence of a problem such as crosstalk due to a layout in which the wiring line connecting the semiconductor substrates is close to the wiring line in a pixel circuit.

(6) Therefore, the present disclosure proposes a solid-state imaging element capable of suppressing crosstalk across wiring lines and reducing noise in a pixel substrate.

Solution to Problem

(7) A solid-state imaging element according to the present disclosure includes: a first semiconductor substrate having a first semiconductor circuit on a first surface of the substrate; a second semiconductor substrate having a second semiconductor circuit on a second surface of the substrate; and a pixel substrate having a pixel circuit on one surface of the substrate, wherein the first semiconductor substrate, the second semiconductor substrate, and the pixel substrate are joined to each other such that the first surface of the first semiconductor substrate and the second surface of the second semiconductor substrate face the one surface of the pixel substrate, and the first semiconductor circuit and the second semiconductor circuit are connected to each other on the first surface side and the second surface side, opposite to the side facing the pixel substrate.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIGS. 1A and 1B are schematic diagrams illustrating a part of a solid-state imaging element according to a first embodiment of the present disclosure.
- (2) FIGS. 2A, 2B, 2C, and 2D are flowcharts illustrating an example of a manufacturing process procedure of the solid-state imaging element according to the first embodiment of the present disclosure.
- (3) FIGS. 3A, 3B, and 3C are flowcharts illustrating the example of the manufacturing process procedure of the solid-state imaging element according to the first embodiment of the present disclosure.
- (4) FIGS. 4A, 4B, and 4C are flowcharts illustrating the example of the manufacturing process procedure of the solid-state imaging element according to the first embodiment of the present disclosure.
- (5) FIGS. 5A and 5B are flowcharts illustrating the example of the manufacturing process procedure of the solid-state imaging element according to the first embodiment of the present disclosure.
- (6) FIGS. 6A and 6B are flowcharts illustrating the example of the manufacturing process procedure of the solid-state imaging element according to the first embodiment of the present disclosure.
- (7) FIG. 7 is a flowchart illustrating an overall image of the manufacturing process of the solid-state imaging element according to the first embodiment of the present disclosure.
- (8) FIG. 8 is a schematic diagram illustrating a part of a solid-state imaging element according to a comparative example of the present disclosure.
- (9) FIGS. 9A, 9B, 9C, and 9D are flowcharts illustrating an example of a manufacturing process procedure of a solid-state imaging element according to a first modification of the first embodiment of the present disclosure.
- (10) FIGS. 10A and 10B are flowcharts illustrating the example of the manufacturing process procedure of the solid-state imaging element according to the first modification of the first embodiment of the present disclosure.
- (11) FIG. 11 is a schematic diagram illustrating a part of a solid-state imaging element according to a second modification of the first embodiment of the present disclosure.
- (12) FIG. 12 is a schematic diagram illustrating a part of a solid-state imaging element according to a third modification of the first embodiment of the present disclosure.
- (13) FIGS. 13A, 13B, and 13C are flowcharts illustrating an example of a manufacturing process procedure of the solid-state imaging element according to the third modification of the first embodiment of the present disclosure.
- (14) FIGS. 14A, 14B, and 14C are flowcharts illustrating the example of the manufacturing process procedure of the solid-state imaging element according to the third modification of the first embodiment of the present disclosure.
- (15) FIG. 15 is a schematic diagram illustrating a part of a solid-state imaging element according to a fourth modification of the first embodiment of the present disclosure.
- (16) FIGS. 16A, 16B, and 16C are schematic diagrams illustrating a part of a solid-state imaging element according to a fifth modification of the first embodiment of the present disclosure.
- (17) FIG. 17 is a schematic diagram illustrating a part of a solid-state imaging element according to a second embodiment of the present disclosure.
- (18) FIGS. 18A and 18B are flowcharts illustrating an example of a manufacturing process procedure of the solid-state imaging element according to the second embodiment of the present disclosure.
- (19) FIGS. 19A and 19B are flowcharts illustrating the example of the manufacturing process procedure of the solid-state imaging element according to the second embodiment of the present disclosure.

disclosure.

(20) FIGS. **20A** and **20B** are flowcharts illustrating the example of the manufacturing process procedure of the solid-state imaging element according to the second embodiment of the present disclosure.

(21) FIGS. **21A** and **21B** are flowcharts illustrating the example of the manufacturing process procedure of the solid-state imaging element according to the second embodiment of the present disclosure.

(22) FIG. **22** is a flowchart illustrating an overall image of the manufacturing process of the solid-state imaging element according to the second embodiment of the present disclosure.

(23) FIG. **23** is a schematic diagram illustrating a part of a solid-state imaging element according to a first modification of the second embodiment of the present disclosure.

(24) FIG. **24** is a schematic diagram illustrating a part of a solid-state imaging element according to a second modification of the second embodiment of the present disclosure.

DESCRIPTION OF EMBODIMENTS

(25) Embodiments of the present disclosure will be described below in detail with reference to the drawings. In each of the following embodiments, the same parts are denoted by the same reference symbols, and a repetitive description thereof will be omitted.

First Embodiment

(26) A solid-state imaging element of a first embodiment will be described with reference to FIGS. **1A**, **1B**, **2A**, **2B**, **2C**, **2D**, **3A**, **3B**, **3C**, **4A**, **4B**, **4C**, **5A**, **5B**, **6A**, **6B**, **7**, **8**, **9A**, **9B**, **9C**, **9D**, **10A**, **10B**, **11**, **12**, **13A**, **13B**, **13C**, **14A**, **14B**, **14C**, **15**, **16A**, **16B**, and **16C**.

Configuration Example of Solid-State Imaging Element

(27) FIGS. **1A** and **1B** are schematic diagrams illustrating a part of a solid-state imaging element **1** according to the first embodiment of the present disclosure. FIG. **1A** is a cross-sectional view illustrating a part of the solid-state imaging element **1** according to the first embodiment; FIG. **1B** is a plan view illustrating the part of the solid-state imaging element **1** according to the first embodiment as viewed from a support substrate **40** side. Note that FIG. **1B** omits illustration of the support substrate **40**.

(28) As illustrated in FIGS. **1A** and **1B**, the solid-state imaging element **1** includes a logic substrate **10** as a first semiconductor substrate, a memory substrate **20** as a second semiconductor substrate, and a pixel substrate **30**. A surface **1A** as a first surface of the logic substrate **10** and a surface **2A** as a second surface of the memory substrate **20** are arranged to face a surface **3A** as one surface of the pixel substrate **30**. The logic substrate **10**, the memory substrate **20**, and the pixel substrate **30** are joined at joining surfaces **JS13** and **JS23** on their facing surface sides. The logic substrate **10** and the memory substrate **20** are further joined to the support substrate **40**, on the side opposite to the side facing the pixel substrate **30**. Examples of the logic substrate **10**, the memory substrate **20**, the pixel substrate **30**, and the support substrate **40** include silicon substrates, or the like.

(29) The pixel substrate **30** includes a pixel circuit **31** including members such as wiring lines **D31** and **D32**, and pixel transistors (not illustrated), on the surface **3A** facing the logic substrate **10** and the memory substrate **20**. The pixel circuit **31** is stacked in an insulating layer **33** on the pixel substrate **30**. Examples of the wiring lines **D31** and **D32** include Cu wiring lines, or the like. The pixel substrate **30** includes a photoelectric conversion element (not illustrated) in the substrate. The pixel substrate **30** includes a color filter CF and an on-chip lens OCL on a surface **3B** opposite to the surface **3A**. The on-chip lens OCL collects emitted light, and then, the collected light is guided to the photoelectric conversion element through the color filter CF. Using photoelectric conversion, the photoelectric conversion element converts the received light into an electric signal corresponding to the amount of received light. The pixel circuit **31** reads out an electric signal from the photoelectric conversion element and outputs the electric signal to the logic substrate **10** side.

(30) The logic substrate **10** includes a logic circuit **11** as a first semiconductor circuit including wiring lines **D11** and **D12**, a logic element (not illustrated), and the like, on the surface **1A** facing a

pixel substrate **30**. The logic circuit **11** is stacked in the insulating layer **13** on the logic substrate **10**. Examples of the wiring lines **D11** and **D12** include Cu wiring lines, or the like. The logic circuit **11** processes the electric signal output from the pixel substrate **30**. The logic circuit **11** is electrically connected to the pixel circuit **31** through an electrode pad **P3** connected to the wiring line **D32** included in the pixel circuit **31** and through an electrode pad **P1** connected to the wiring line **D12** included in the logic circuit **11**. Examples of the material of the electrode pads **P1** and **P3** include Cu.

(31) The memory substrate **20** includes a memory circuit **21** as a second semiconductor circuit including wiring lines **D21** and **D22**, a storage element (not illustrated), and the like, on the surface **2A** facing the pixel substrate **30**. The memory circuit **21** is stacked in an insulating layer **23** on the memory substrate **20**. Examples of the wiring lines **D21** and **D22** include Cu wiring lines, or the like. The memory circuit **21** holds various data necessary for the solid-state imaging element **1** to function. The memory circuit **21** is electrically connected to the pixel circuit **31** through the electrode pad **P3** connected to the wiring line **D32** included in the pixel circuit **31** and through an electrode pad **P2** connected to the wiring line **D22** included in the memory circuit **21**. Examples of the material of the electrode pad **P2** include Cu.

(32) The logic circuit **11** and the memory circuit **21** are connected to each other on the side of surfaces **1B** and **2B**, opposite to the side facing the pixel substrate **30**.

(33) More specifically, the logic substrate **10** and the memory substrate **20** are connected to each other through a via **12** as a first via penetrating the logic substrate **10**, a via **22** as a second via penetrating the memory substrate **20**, and a wiring line **DD** in a wiring layer connecting the vias **12** and **22** to each other. Examples of the material of the vias **12** and **22** and the wiring line **DD** include Cu. The vias **12** and **22** are Through Silicon Vias (TSVs) penetrating through the logic substrate **10** and the memory substrate **20**, respectively. The vias **12** and **22** have, for example, reverse tapered shapes in which the diameter increases from the surfaces **1A** and **2A** side to the surfaces **1B** and **2B** side. As described below, a wiring layer in which the wiring line **DD** is disposed is a redistribution layer (RDL) formed by a plating technique, for example. The wiring line **DD** is arranged on the surface **1B** of the logic substrate **10**, the surface **2B** of the memory substrate **20**, and an insulating layer **53** filling the space between the logic substrate **10** and the memory substrate **20**.

(34) As illustrated in FIG. **1B**, the logic substrate **10** and the memory substrate **20** may be connected to each other by a plurality of wiring lines **DD**. The plurality of wiring lines **DD** may be arranged in parallel to each other, for example.

(35) The support substrate **40** has: a surface **4A** facing the surface **1B** of the logic substrate **10** and the surface **2B** of the memory substrate **20**; and a surface **4B** on a side opposite to the surface **4A**. The support substrate **40** is joined to the logic substrate **10** and the memory substrate **20** through an insulating layer **43** at joining surfaces **JS14** and **JS24**.

Example of Process of Manufacturing Solid-State Imaging Element

(36) Next, an example of a process of manufacturing the solid-state imaging element **1** according to the first embodiment will be described with reference to FIGS. **2A**, **2B**, **20**, **2D**, **3A**, **3B**, **3C**, **4A**, **4B**, **4C**, **5A**, **5B**, **6A**, **6B**, and **7**. FIGS. **2A**, **2B**, **2C**, **2D**, **3A**, **3B**, **3C**, **4A**, **4B**, **4C**, **5A**, **5B**, **6A**, and **6B** are flowcharts illustrating an example of a manufacturing process procedure of the solid-state imaging element **1** according to the first embodiment of the present disclosure.

(37) As illustrated in FIG. **2A**, the memory substrate **20**, in which the memory circuit **21** is stacked in the insulating layer **23**, is formed. After the formation of the wiring lines **D21** and **D22**, the electrode pad **P2**, and the like, the memory substrate **20** is ground to be thinned.

(38) As illustrated in FIG. **2B**, the logic substrate **10**, in which the logic circuit **11** is stacked in the insulating layer **13**, is formed. After the formation of the wiring lines **D11** and **D12**, the electrode pad **P1**, and the like, the logic substrate **10** is ground to be thinned.

(39) As illustrated in FIG. **2C**, the pixel substrate **30**, in which the pixel circuit **31** is stacked in the insulating layer **33**, is formed. The pixel substrate **30** is provided with the wiring lines **D31** and

D32, the electrode pad P3, and the like. The pixel substrate 30 may also be ground to be thinned.

(40) The order of formation of the logic substrate 10, the memory substrate 20, and the pixel substrate 30 is not limited.

(41) As illustrated in FIG. 2D, the logic substrate 10 and the memory substrate 20 are bonded to the pixel substrate 30 such that the surface 1A of the logic substrate 10 and the surface 2A of the memory substrate 20 face the surface 3A of the pixel substrate 30.

(42) As illustrated in FIG. 3A, using the procedure described above, the logic substrate 10 is joined to the pixel substrate 30 at the joining surface JS13 through the electrode pads P1 and P3. In addition, the memory substrate 20 is joined to the pixel substrate 30 at the joining surface JS23 through the electrode pads P2 and P3. The electrode pads P1 and P3 are joined to each other and the electrode pads P2 and P3 are joined to each other using Cu—Cu joining, for example.

(43) Using a chemical vapor deposition (CVD) process, for example, the insulating layer 53 filling the space between the logic substrate 10 and the memory substrate 20 is formed, and then, the insulating layer 53 is planarized by a chemical mechanical polishing (CMP) process, for example.

(44) As illustrated in FIG. 3B, a through hole H12 is formed to penetrate the logic substrate 10 from the surface 1B side to reach the wiring line D11. In addition, a through hole H22 is formed to penetrate the memory substrate 20 from the surface 2B side to reach the wiring line D21. These through holes 12 and 22 tend to have a tapered shape in which the diameter decreases from the surfaces 1B and 2B side to the surfaces 1A and 2A side.

(45) As illustrated in FIG. 3C, the through hole H12 is filled with a conductive material such as Cu to form the via 12 that penetrates the logic substrate 10 to be connected to the wiring line D11, for example. In addition, the through hole H22 is filled with a conductive material such as Cu to form the via 22 that penetrates the memory substrate 20 to be connected to the wiring line D21, for example. For example, by having a tapered shape in which the diameter decreases from the surfaces 1B and 2B side to the surfaces 1A and 2A side, the vias 12 and 22 can be determined to have been formed from the surfaces 1B and 2B side.

(46) As illustrated in FIG. 4A, a resin 60 is formed on the surface 1B of the logic substrate 10 and the surface 2B of the memory substrate 20. The resin 60 has a trench pattern 60tr in a region including the via 12 of the logic substrate 10 and the via 22 of the memory substrate 20.

(47) As illustrated in FIG. 4B, the trench pattern 60tr in the resin 60 is filled with a conductive material such as Cu by a plating method, for example, so as to form the wiring line DD.

(48) As illustrated in FIG. 4C, the resin 60 is peeled off. This forms the wiring line DD in the redistribution layer. For formation of the redistribution layer, an insulating layer having a trench pattern, or the like, can be used instead of the resin 60 having the trench pattern 60tr.

(49) As illustrated in FIG. 5A, the insulating layer 43 is formed on the surface 1B of the logic substrate 10 and the surface 2B of the memory substrate 20 so as to cover the wiring line DD.

(50) As illustrated in FIG. 5B, the logic substrate 10 and the memory substrate 20 are bonded to the support substrate 40 such that the surface 1B of the logic substrate 10 and the surface 2B of the memory substrate 20 face the surface 4A of the support substrate 40.

(51) As illustrated in FIG. 6A, using the procedure described above, the logic substrate 10 is joined to the support substrate 40 at the joining surface JS14 through the insulating layer 43. In addition, the memory substrate 20 is joined to the support substrate 40 at a joining surface JS24 through the insulating layer 43.

(52) As illustrated in FIG. 6B, the color filter CF and the on-chip lens OCL are formed on the surface 3B of the pixel substrate 30.

(53) FIG. 7 illustrates an overall flow of the process of manufacturing the solid-state imaging element 1 according to the first embodiment. FIG. 7 is a flowchart illustrating an overall image of a manufacturing process of the solid-state imaging element 1 according to the first embodiment of the present disclosure.

(54) As illustrated in FIG. 7, the pixel circuit 31, the logic circuit 11, and the memory circuit 21 are

formed by the front surface process on the pixel substrate **30**, the logic substrate **10**, and the memory substrate **20**, respectively. Thereafter, the back surfaces of the logic substrate **10** and the memory substrate **20**, namely, the surfaces **1B** and **2B**, are ground.

(55) Next, the logic substrate **10** and the memory substrate **20** are bonded to the pixel substrate **30**.
(56) Next, using a back surface process of the logic substrate **10** and the memory substrate **20**, the vias **12** and **22** as TSVs are formed in the logic substrate **10** and the memory substrate **20**, respectively, and further, the wiring line DD in the redistribution layer connecting these vias **12** and **22** is formed.

(57) Next, the logic substrate **10** and the memory substrate **20**, which have been joined to the pixel substrate **30**, are bonded to the support substrate **40**.

(58) Next, the color filter CF and the on-chip lens OCL are formed on the pixel substrate **30** by the back surface process of the pixel substrate **30**.

(59) The above-described procedure completes the process of manufacturing the solid-state imaging element **1** according to the first embodiment.

Comparative Example

(60) Next, problems of a solid-state imaging element of a comparative example will be described with reference to FIG. **8**. As illustrated in FIG. **8**, the solid-state imaging element of the comparative example has a configuration in which a logic substrate **10'** having a logic circuit **11'** and a memory substrate **20'** having a memory circuit **21'** are joined to a pixel substrate **30'** having a pixel circuit **31'**. These substrates **10'**, **20'**, and **30'** are further supported by a support substrate **40'**. The logic circuit **11'** and the memory circuit **21'** are electrically connected to each other through electrode pads **P1'** and **P3'**, a wiring line DD', and electrode pads **P3'** and **P2'**, on the joining surface side with the pixel substrate **30'**. The wiring line DD' is formed on the pixel substrate **30'** side.

(61) Unfortunately, however, such a connection method has a problem of proximity in which a distance X' between the wiring line DD' and a wiring line D31' of the pixel substrate **30'** is too short, leading to a concern of occurrence of crosstalk. Such crosstalk would cause occurrence of noise in the pixel circuit **31'**, leading to degradation of the characteristics of the solid-state imaging element. Another problem is that the wiring line DD' is formed in the region of the logic circuit **11'**, causing a restriction on the layout in the wiring design. In a case where a shield is applied to the wiring line DD' in order to suppress crosstalk, the restriction regarding the layout would be more severe.

(62) In contrast, according to the solid-state imaging element **1** of the first embodiment, the logic circuit **11** and the memory circuit **21** are connected to each other on the side of the surfaces **1B** and **2B**, opposite to the side facing the pixel substrate **30**. With this configuration, the wiring line DD connecting the logic circuit **11** and the memory circuit **21** to each other and the wiring line D31 and the like of the pixel substrate **30** are separated from each other by a distance which is the thickness of the logic substrate **10** and the memory substrate **20**, or more. This makes it possible to suppress the crosstalk between the wiring line DD and the wiring line D31, leading to achievement of noise reduction in the pixel circuit **31**.

(63) According to the solid-state imaging element **1** of the first embodiment, the wiring line DD is arranged on the surface **1B** side of the logic substrate **10** and the surface **2B** side of the memory substrate **20**. This eliminates the necessity to connect the logic circuit **11** and the memory circuit **21** to each other using the region of the logic circuit **11**, increasing the degree of freedom of layout in the wiring design.

(64) According to the solid-state imaging element **1** of the first embodiment, the wiring layer including the wiring line DD connecting the logic circuit **11** and the memory circuit **21** to each other is a redistribution layer formed by a plating technique, for example. This makes it possible to form the wiring line DD by a simple and inexpensive method.

First Modification

(65) Next, a solid-state imaging element according to a first modification of the first embodiment

will be described with reference to FIGS. 9A, 9B, 9D, 10A, and 10B. The solid-state imaging element of the first modification is different from that of the first embodiment in a part of the manufacturing process procedure.

(66) As illustrated in FIG. 9A, the memory substrate **20**, in which the memory circuit **21** is stacked in the insulating layer **23**, is formed. The memory substrate **20** is not ground at this stage. The memory substrate **20** is provided with a via **22a** reaching halfway through a thick substrate. The via **22a** is formed from the surface **2A** side of the memory substrate **20** (front surface process), and the wiring line **D21** is thereafter formed so as to be connected to the upper surface of the via **22a**.

(67) As illustrated in FIG. 9B, the logic substrate **10**, in which the logic circuit **11** is stacked in the insulating layer **13**, is formed. The logic substrate **10** is not ground at this stage. The logic substrate **10** is provided with a via **12a** reaching halfway through the thick substrate. The via **12a** is formed from the surface **1A** side of the logic substrate **10** (front surface process), and the wiring line **D11** is thereafter formed so as to be connected to the upper surface of the via **12a**.

(68) As illustrated in FIG. 9C, the pixel substrate **30**, in which the pixel circuit **31** is stacked in the insulating layer **33**, is formed. Formation of the logic substrate **10**, the memory substrate **20**, and the pixel substrate **30** may be performed in any order.

(69) As illustrated in FIG. 9D, the logic substrate **10** and the memory substrate **20** are bonded to the pixel substrate **30** such that the surface **1A** of the logic substrate **10** and the surface **2A** of the memory substrate **20** face the surface **3A** of the pixel substrate **30**.

(70) The insulating layer **53** filling a space between the logic substrate **10** and the memory substrate **20** is formed by the CVD process, for example, and the insulating layer **53** is planarized by the CMP process, for example.

(71) The logic substrate **10**, the memory substrate **20**, and the insulating layer **50** therebetween are ground, so as to thin the logic substrate **10** and the memory substrate **20** until the end portions of the vias **12a** and **22a** on the surfaces **1B** and **2B** sides are exposed.

(72) Thereafter, the solid-state imaging element of the first modification is manufactured in a procedure similar to that in the above-described first embodiment.

(73) That is, as illustrated in FIG. 10A, the wiring line **DD** connected to the end portions of the vias **12a** and **22a** on the surfaces **1B** and **2B** sides is formed by using a redistribution technique, for example. Furthermore, as illustrated in FIG. 10B, the logic substrate **10** and the memory substrate **20**, to which the pixel substrate **30** has been joined, are to be joined to the support substrate **40**, so as to form the color filter **CF** and the on-chip lens **OCL** on the pixel substrate **30**.

(74) The above-described procedure completes the process of manufacturing the solid-state imaging element according to the first modification. In the solid-state imaging element of the first modification, the via **12a**, which is the first via formed from the surface **1A** side, has a tapered shape in which the diameter decreases from the surface **1A** side to the surface **1B** side, for example. Furthermore, the via **22a**, which is the second via formed from the surface **2A** side, has a tapered shape in which the diameter decreases from the surface **2A** side to the surface **2B** side, for example.

Second Modification

(75) Next, a solid-state imaging element **1b** according to a second modification of the first embodiment will be described with reference to FIG. 11. The solid-state imaging element **1b** of the second modification is different from the case of the first embodiment in that the wiring layer in which a wiring line **DDb** is arranged is not a redistribution layer.

(76) As illustrated in FIG. 11, the solid-state imaging element **1b** includes the wiring line **DDb** stacked in the insulating layer **43**. More specifically, the wiring line **DDb** has two via portions **VI** and one bridge portion **CR**. One end of one via portion **VI** is connected to an end portion of the via **12** of the logic substrate **10** on the surface **1B** side, while the other end is connected to one end of the bridge portion **CR**. One end of the other via portion **VI** is connected to an end portion on the surface **2B** side of the via **22** of the memory substrate **20**, and the other end is connected to the other end of the bridge portion **CR**. The wiring line **DDb** having such a structure is formed by a

damascene process using a photolithography technique and a dry etching technique, for example. (77) According to the solid-state imaging element **1b** of the second modification, the wiring line DDb is formed by a photolithography technique and a dry etching technique. This makes it possible to form a finer wiring line DDb with high accuracy, leading to achievement of high integration of the wiring line DDb.

Third Modification

(78) Next, a solid-state imaging element **1c** according to a third modification of the first embodiment will be described with reference to FIGS. **12**, **13A**, **13B**, **13C**, **14A**, **14B**, and **14C**. In addition to the wiring line DDb, the solid-state imaging element **1c** of the third modification further includes a shield SSc to which a method of forming the wiring line DDb is applied.

(79) As illustrated in FIG. **12**, the shield SSc disposed in a shield layer as a conductive layer is arranged between the joining surfaces JS**13** and JS**23**, which join the pixel substrate **30** with the logic substrate **10** and the memory substrate **20**, and the wiring line DDb. Similarly to the wiring line DDb, the shield SSc is also stacked in the insulating layer **43**. More specifically, the shield SSc includes two via portions V**1c** and one bridge portion CR**c**. One end of one via portion V**1c** is connected to the surface **1B** of the logic substrate **10**, while the other end is connected to one end of the bridge portion CR**c**. One end of the other via portion V**1c** is connected to the surface **2B** of the memory substrate **20**, while the other end is connected to the other end of the bridge portion CR**c**. Similarly to the wiring line DDb, the shield SSc having such a structure is formed by the damascene process using a photolithography technique and a dry etching technique, for example.

(80) The wiring line DDb and the shield SSc can be formed using, for example, the dual damascene process illustrated in FIGS. **13A**, **13B**, **13C**, **14A**, **14B**, and **14C**. The flow illustrated from FIGS. **13A**, **13B**, **13C**, **14A**, **14B**, and **14C** are replacements for the flow of FIGS. **4A**, **4B**, and **4C** of the first embodiment described above. FIGS. **13A**, **13B**, **13C**, **14A**, **14B**, and **14C** are enlarged illustrations of the vicinity of the end portion of the via **12** of the logic substrate **10** on the surface **1B** side and the end portion of the via **22** of the memory substrate **20** on the surface **2B** side.

(81) As illustrated in FIG. **13A**, the insulating layer **43** is formed on the surface **1B** of the logic substrate **10**, on the insulating layer **53**, and on the surface **2B** of the memory substrate **20**. At this time, the insulating layer **43** is formed to be thinner than the final thickness.

(82) As illustrated in FIG. **13B**, for example, a photolithography technique and a dry etching technique are used to form holes HL**c**, each of which penetrating the insulating layer **43** to reach the surface **1B** of the logic substrate **10** and the surface **2B** of the memory substrate **20**, individually, and to form a trench TR**c** having each end portion connected to each of the holes HL**c**.

(83) As illustrated in FIG. **13C**, the inside of the hole HL**c** and the inside of the trench TR**c** are filled with a conductive material such as Cu by the CVD process, for example, and an excessive conductive material is removed by the CMP process or the like to form the shield SSc having the via portions V**1c** and the bridge portion CR**c**.

(84) As illustrated in FIG. **14A**, the insulating layer **43** is further formed so as to cover the shield SSc.

(85) As illustrated in FIG. **14B**, a photolithography technique and a dry etching technique are used, for example, to form holes HL**b**, each of which penetrating the insulating layer **43** to reach the via **12** of the logic substrate **10** and the via **22** of the memory substrate **20**, individually, and to form a trench TR**b** having each end portion connected to each of the holes HL**b**.

(86) As illustrated in FIG. **14C**, the inside of the hole HL**b** and the inside of the trench TR**b** are filled with a conductive material such as Cu by the CVD process, for example, and an excessive conductive material is removed by the CMP process or the like to form the wiring line DDb having the via portions V**1** and the bridge portion CR.

(87) The above-described procedure completes the formation of the wiring line DDb and the shield SSc. In the above example, the dual damascene process of collectively forming the via portions and the bridge portion is used, but a single damascene process of separately forming the via portions

and the bridge portion may be used.

(88) According to the solid-state imaging element **1c** of the third modification, the shield SSc is disposed on the pixel substrate **30** side of the wiring line DDb. With this configuration, the wiring line DDb is shielded against the wiring line D31 and the like of the pixel substrate **30**, making it possible to further suppress the crosstalk between the wiring line DDb and the wiring line D31, leading to achievement of the reduction of the noise in the pixel circuit **31**.

Fourth Modification

(89) Next, a solid-state imaging element **1d** of a fourth modification of the first embodiment will be described with reference to FIG. **15**. The solid-state imaging element **1d** of the fourth modification is different from the case of the first embodiment described above in that two wiring lines DDda and DDdb intersect each other.

(90) As illustrated in the plan view of FIG. **15(a)**, the wiring lines DDda and DDdb connecting the logic substrate **10** and the memory substrate **20** to each other intersect each other.

(91) As illustrated in the perspective view of FIG. **15(b)**, such a configuration can be obtained by arranging one of the wiring lines, namely, the wiring line DDdb, to be positioned higher than the other wiring line, namely the wiring line DDda, so as to allow the two wiring lines DDda and DDdb to three-dimensionally intersect each other.

(92) These wiring lines DDda and DDdb can be easily formed by the photolithography technique and the dry etching technique as described in the above second and third modifications, for example. The wiring layer including the one of the wiring lines, namely, the wiring line DDda, may be a redistribution layer using the plating technique as described in the first embodiment.

Alternatively, vias may be further formed on the vias **12** and **22** so as to form both wiring layers including the wiring lines DDda and DDdb are to be implemented as redistribution layers using a plating technique.

Fifth Modification

(93) Although the above-described first embodiment and the like have a configuration in which one logic substrate **10** and one memory substrate **20** are joined to the pixel substrate **30**, the present invention is not limited to this configuration. Hereinafter, some examples of joining various different substrates to the pixel substrate **30** will be described with reference to FIGS. **16A**, **16B**, and **16C** as a solid-state imaging element of a fifth modification of the first embodiment.

(94) As illustrated in FIG. **16A**, a solid-state imaging element **1e** has a configuration in which two logic substrates **10** and **10e** are bonded to a pixel substrate **30**. The logic circuits included in the two logic substrates **10** and **10e** are electrically connected to each other by a via **12** of the logic substrate **10**, a via **10e** of the logic substrate **10e**, and a wiring line DDe joining these vias **12** and **10e** to each other.

(95) As illustrated in FIG. **16B**, a solid-state imaging element **1f** has a configuration in which two memory substrates **20** and **20f** are bonded to a pixel substrate **30**. The memory circuits included in the two memory substrates **20** and **20f** are electrically connected to each other by a via **22** of the memory substrate **20**, a via **20f** of the memory substrate **20f**, and a wiring line DDf joining these vias **22** and **20f** to each other.

(96) As illustrated in FIG. **16C**, a solid-state imaging element **1g** has a configuration in which one logic substrate **10** and two memory substrates **20ga** and **20gb** are bonded to a pixel substrate **30**. The logic circuit of the logic substrate **10** and the memory circuit of the memory substrate **20ga** are electrically connected to each other by a via **12** of the logic substrate **10**, a via **22ga** of the memory substrate **20ga**, and a wiring line DDga joining these vias **12** and **22ga** to each other. The logic circuit of the logic substrate **10** and the memory circuit of the memory substrate **20gb** are electrically connected to each other by a via **12** of the logic substrate **10**, a via **22gb** of the memory substrate **20gb**, and a wiring line DDgb joining these vias **12** and **22gb** to each other. The memory circuits included in the memory substrate **20ga** and the memory substrate **20gb** are electrically connected to each other by the via **22ga** of the memory substrate **20ga**, the via **22gb** of the memory

substrate **20gb**, and a wiring line DDgab joining these vias **22ga** and **22gb** to each other.

(97) In addition to the above, the solid-state imaging element of the fifth modification can have a configuration in which various substrates of arbitrary numbers are bonded with the pixel substrate **30**. The solid-state imaging element of the fifth modification may include a shield layer in addition to the wiring layer.

Second Embodiment

(98) Next, a solid-state imaging element **2** of a second embodiment will be described with reference to FIGS. **17**, **18A**, **18B**, **19A**, **19B**, **20A**, **20B**, **21A**, **21B**, **22**, **23**, and **24**. The solid-state imaging element **2** of the second embodiment is different from the case of the first embodiment described above in that a wiring line DDh is provided on the surface **4B** side of the support substrate **40**.

Configuration Example of Solid-State Imaging Element

(99) FIG. **17** is a schematic diagram illustrating a part of the solid-state imaging element **2** according to the second embodiment of the present disclosure. As illustrated in

(100) FIG. **17**, the solid-state imaging element **2** includes the support substrate **40** directly joined both to the logic substrate **10** and to the memory substrate **20**. That is, the surface **1B** of the logic substrate **20** and the surface **2B** of the memory substrate **20** are bonded to the surface **4A** of the support substrate **40** at joining surfaces JS**14h** and JS**24h**, respectively.

(101) The logic circuit **11** and the memory circuit **21** are electrically connected to each other through a via **12h** as a first via penetrating the support substrate **40** and the logic substrate **10**, a via **22h** as a second via penetrating the support substrate **40** and the memory substrate **20**, and the wiring line DDh joining these vias **12h** and **22h** to each other.

(102) The wiring line DDh is disposed on the surface **4B** of the support substrate **40**, which is the surface on the side opposite to the side on which the logic substrate **10** and the memory substrate **20** are disposed. On the surface **4B** of the support substrate **40**, an insulating layer **73** is disposed to cover the wiring line DDh.

Example of Process of Manufacturing Solid-State Imaging Element

(103) Next, an example of a process of manufacturing the solid-state imaging element **2** according to the second embodiment will be described with reference to FIGS. **18A**, **18B**, **19A**, **19B**, **20A**, **20B**, **21A**, **21B**, and **22**. FIGS. **18A**, **18B**, **19A**, **19B**, **20A**, **20B**, **21A**, and **21B** are flowcharts illustrating an example of a manufacturing process procedure of the solid-state imaging element **2** according to the second embodiment of the present disclosure.

(104) As illustrated in FIG. **18A**, the pixel substrate **30** in which the logic substrate **10** and the memory substrate **20** are bonded to each other is formed. The process up to this point is similar to the process up to FIG. **3A** of the first embodiment described above.

(105) As illustrated in FIG. **18B**, the surface **4A** of the support substrate **40** is bonded to the surface **1B** of the logic substrate **10** and the surface **2B** of the memory substrate **20**, which have been bonded to the pixel substrate **30**.

(106) As illustrated in FIG. **19A**, a through hole H**12h** which penetrates the support substrate **40** and the logic substrate **10** from the surface **4B** side of the support substrate **40** and reaches, for example, a wiring line DD**11** is formed. In addition, a through hole H**22h** which penetrates the support substrate **40** and the memory substrate **20** from the surface **4B** side of the support substrate **40** and reaches, for example, a wiring line DD**21** is formed.

(107) As illustrated in FIG. **19B**, the through hole H**12h** is filled with a conductive material such as Cu to form the via **12h** that penetrates the support substrate **40** and the logic substrate **10** to be connected, for example, to the wiring line D**11**. In addition, the through hole H**22h** is filled with a conductive material such as Cu to form the via **22h** which penetrates the support substrate **40** and the memory substrate **20** to be connected, for example, to the wiring line D**21**.

(108) The vias **12h** and **22h** formed from the surface **4B** side of the support substrate **40** each have a tapered shape in which the diameter decreases from the surface **4B** side of the support substrate

40 to the surface **1A** side of the logic substrate **10** and the surface **2A** side of the memory substrate **20**, for example.

(109) As illustrated in FIG. **20A**, a resin **60h** having a trench pattern **60htr** is formed in a region including the vias **12h** and **22h** on the surface **4B** of the support substrate **40**. Subsequently, the trench pattern **60htr** is filled with a conductive material such as Cu by a plating method, for example, to form the wiring line DDh.

(110) As illustrated in FIG. **20B**, after the resin **60h** is peeled off, the insulating layer **73** is formed on the surface **4B** of the support substrate **40** so as to cover the wiring line DDh.

(111) As illustrated in FIG. **21A**, the stacked substrates, namely, the support substrate **40**, logic substrate **10**, memory substrate **20**, and pixel substrate **30**, are inverted, to be arranged such that the support substrate **40** faces downward and the pixel substrate **30** faces upward.

(112) As illustrated in FIG. **21B**, the color filter CF and the on-chip lens OCL are formed on the surface **3B** of the pixel substrate **30**.

(113) FIG. **22** illustrates an overall flow of manufacturing process of the solid-state imaging element **2** of the second embodiment. FIG. **22** is a flowchart illustrating an overall image of a manufacturing process of the solid-state imaging element **2** according to the second embodiment of the present disclosure.

(114) As illustrated in FIG. **22**, the process is similar to that of the first embodiment described above up to the point where the logic substrate **10** and the memory substrate **20** are bonded to the pixel substrate **30**.

(115) Next, the support substrate **40** is joined to the logic substrate **10** and the memory substrate **20**, which have been bonded to the pixel substrate **30**.

(116) Next, using a back surface process of the support substrate **40**, the vias **12h** and **22h** as TSVs are formed in the support substrate **40**, the logic substrate **10** and the memory substrate **20**, respectively, and further, the wiring line DDh in the redistribution layer connecting these vias **12h** and **22h** is formed on the support substrate **40**.

(117) Next, the color filter CF and the on-chip lens OCL are formed on the pixel substrate **30** by the back surface process of the pixel substrate **30**.

(118) The above-described procedure completes the process of manufacturing the solid-state imaging element **2** according to the second embodiment.

(119) According to the solid-state imaging element **1** of the first embodiment, the logic circuit **11** and the memory circuit **21** are connected to each other on the surface **4B** side of the support substrate **40**. With this configuration, the wiring line DDh connecting the logic circuit **11** and the memory circuit **21** to each other and the wiring line D31 and the like of the pixel substrate **30** are separated by a distance which is a sum of the thickness of the logic substrate **10**, the memory substrate **20**, and the thickness of the support substrate **40**, or more. This makes it possible to further suppress the crosstalk between the wiring line

(120) DD and the wiring line D31, leading to enhanced noise reduction in the pixel circuit **31**.

First Modification

(121) Next, a solid-state imaging element **2i** according to a first modification of the second embodiment will be described with reference to FIG. **23**. The solid-state imaging element **2i** of the first modification includes a wiring line DDi formed by a photolithography technique and a dry etching technique.

(122) FIG. **23** is a schematic diagram illustrating a part of the solid-state imaging element **2i** according to the first modification of the second embodiment of the present disclosure. As illustrated in FIG. **23**, the solid-state imaging element **2i** includes the wiring line DDi having two via portions and one bridge portion, similarly to the second modification of the first embodiment described above. The wiring line DDi is stacked in the insulating layer **73**. The wiring line DDi is formed by a photolithography technique and a dry etching technique, for example.

(123) According to the solid-state imaging element **2i** of the first modification, the finer wiring line

DDi can be accurately formed by the photolithography technique and the dry etching technique, for example, leading to achievement of high integration of the wiring line DDi.

Second Modification

(124) Next, a solid-state imaging element **2j** according to a second modification of the second embodiment will be described with reference to FIG. **24**. In addition to the wiring line DDi, the solid-state imaging element **2j** of the second modification further includes a shield SSj which is disposed in a shield layer as a conductive layer and to which a method of forming the wiring line DDi is applied.

(125) FIG. **24** is a schematic diagram illustrating a part of a solid-state imaging element **2j** according to the second modification of the second embodiment of the present disclosure. As illustrated in FIG. **24**, the solid-state imaging element **2j** includes the shield SSj disposed in a shield layer having two via portions and one bridge portion, similarly to the third modification of the first embodiment described above. The shield SSj is also stacked in the insulating layer **73**. Similarly to the wiring line DDi, the shield SSj is formed by the photolithography technique and the dry etching technique.

(126) According to the solid-state imaging element **2j** of the second modification, the crosstalk between the wiring line DDb and the wiring line D31 can be further suppressed by the shield SSj arranged on the pixel substrate **30** side of the wiring line DDi, leading to enhanced noise reduction in the pixel circuit **31**.

Third Modification

(127) Even with a configuration having the wiring layer on the support substrate **40** as in the solid-state imaging element of the second embodiment and the like, it is possible to form a configuration in which various substrates of an arbitrary number, such as a plurality of logic substrates and a plurality of memory substrates, are bonded to the pixel substrate **30**, as described as the configuration of the fifth modification of the first embodiment.

(128) The solid-state imaging element of the third modification may include a shield layer in addition to the wiring layer on the support substrate **40**.

(129) The effects described in the present specification are merely examples, and thus, there may be other effects, not limited to the exemplified effects.

Other Embodiments

(130) Note that the present technology can also have the following configurations.

(131) (1)

(132) A solid-state imaging element comprising: a first semiconductor substrate having a first semiconductor circuit on a first surface of the substrate; a second semiconductor substrate having a second semiconductor circuit on a second surface of the substrate; and a pixel substrate having a pixel circuit on one surface of the substrate, wherein the first semiconductor substrate, the second semiconductor substrate, and the pixel substrate are joined to each other such that the first surface of the first semiconductor substrate and the second surface of the second semiconductor substrate face the one surface of the pixel substrate, and the first semiconductor circuit and the second semiconductor circuit are connected to each other on the first surface side and the second surface side, opposite to the side facing the pixel substrate.

(2)

(133) The solid-state imaging element according to (1), wherein the first semiconductor circuit and the second semiconductor circuit are connected to each other through a first via penetrating the first semiconductor substrate, a second via penetrating the second semiconductor substrate, and a wiring line connecting the first via and the second via to each other.

(3)

(134) The solid-state imaging element according to (2), wherein a shield is disposed between a joining surface which joins the first semiconductor substrate and the second semiconductor substrate to the pixel substrate, and the wiring line.

- (4)
(135) The solid-state imaging element according to (2) or (3), wherein the wiring line is disposed on the first semiconductor substrate and the second semiconductor substrate.
- (5)
(136) The solid-state imaging element according to (2) or (3), further comprising a support substrate joined to a side of the first semiconductor substrate and the second semiconductor substrate, opposite to the side facing the pixel substrate, wherein the first via and the second via each penetrate the support substrate, and the wiring line is disposed on the support substrate on a side opposite to the first semiconductor substrate and the second semiconductor substrate.
- (6)
(137) The solid-state imaging element according to any one of (2) to (5), wherein a wiring layer in which the wiring line is disposed is a redistribution layer formed by a plating technique.
- (7)
(138) The solid-state imaging element according to any one of (2) to (5), wherein a wiring layer in which the wiring line is disposed is a wiring layer formed by a photolithography technique and a dry etching technique.
- (8)
(139) The solid-state imaging element according to any one of (1) to (7), wherein the first semiconductor substrate is a logic substrate including a logic circuit as the first semiconductor circuit, and the second semiconductor substrate is a memory substrate including a memory circuit as the second semiconductor circuit.
- (9)
(140) The solid-state imaging element according to any one of (1) to (7), wherein the first semiconductor substrate is a first logic substrate including a first logic circuit as the first semiconductor circuit, and the second semiconductor substrate is a second logic substrate including a second logic circuit as the second semiconductor circuit.
- (10)
(141) The solid-state imaging element according to any one of (1) to (7), wherein the first semiconductor substrate is a first memory substrate including a first memory circuit as the first semiconductor circuit, and the second semiconductor substrate is a second memory substrate including a second memory circuit as the second semiconductor circuit.
- (11)
(142) The solid-state imaging element according to any one of (1) to (7), further comprising a third semiconductor substrate having a third semiconductor circuit on a third surface of the substrate, wherein the third semiconductor substrate and the pixel substrate are joined to each other such that the third surface of the third semiconductor substrate and the one surface of the pixel substrate face each other, the first semiconductor circuit and the third semiconductor circuit are connected to each other on the first surface side and the third surface side, opposite to the side facing the pixel substrate, and the second semiconductor circuit and the third semiconductor circuit are connected to each other on the second surface side and the third surface side, opposite to the side facing the pixel substrate.
- (12)
(143) The solid-state imaging element according to (11), further comprising: a first wiring line that connects the first semiconductor circuit and the third semiconductor circuit to each other; and a second wiring line that connects the second semiconductor circuit and the third semiconductor circuit to each other, wherein the first wiring line is disposed on the first semiconductor substrate and the third semiconductor substrate, and the second wiring line is disposed on the second semiconductor substrate and the third semiconductor substrate.
- (13)
(144) The solid-state imaging element according to (11) or (12), wherein the first semiconductor

substrate is a logic substrate including a logic circuit as the first semiconductor circuit, the second semiconductor substrate is a first memory substrate including a first memory circuit as the second semiconductor circuit, and the third semiconductor substrate is a second memory substrate including a second memory circuit as the third semiconductor circuit.

(14)

(145) The solid-state imaging element according to (5), further comprising: a third semiconductor substrate having a third semiconductor circuit on a third surface of the substrate; a first wiring line that connects the first semiconductor circuit and the third semiconductor circuit to each other on the first surface side and the third surface side, opposite to the side facing the pixel substrate; and a second wiring line that connects the second semiconductor circuit and the third semiconductor circuit to each other on the second surface side and the third surface side, opposite to the side facing the pixel substrate, wherein the third surface of the third semiconductor substrate and the one surface of the pixel substrate are joined to each other so as to face each other, and the first wiring line and the second wiring line are disposed on the support substrate on a side opposite to the third semiconductor substrate.

REFERENCE SIGNS LIST

(146) **1, 1b, 1c, 2, 2i, 2j** SOLID-STATE IMAGING ELEMENT **10** LOGIC SUBSTRATE **11** LOGIC CIRCUIT **12, 12a, 12h** VIA **20** MEMORY SUBSTRATE **21** MEMORY CIRCUIT **22, 22a, 22h** VIA **30** PIXEL SUBSTRATE **31** PIXEL CIRCUIT **DD, DDb, DD, DDh, DDi** WIRING LINE **SSc, SSj** SHIELD

Claims

1. A solid-state imaging element, comprising: a first semiconductor substrate that includes a first semiconductor circuit, wherein the first semiconductor circuit is on a first surface of the first semiconductor substrate; a second semiconductor substrate that includes a second semiconductor circuit, wherein the second semiconductor circuit is on a first surface of the second semiconductor substrate; a pixel substrate that includes a pixel circuit, wherein the pixel circuit is on a surface of the pixel substrate; a joining surface at which the first semiconductor substrate and the second semiconductor substrate are joined to the pixel substrate, wherein each of the first surface of the first semiconductor substrate and the first surface of the second semiconductor substrate faces the surface of the pixel substrate, the first semiconductor circuit is connected to the second semiconductor circuit on a second surface of the first semiconductor substrate and a second surface of the second semiconductor substrate, the second surface of the first semiconductor substrate is opposite to the first surface of the first semiconductor substrate, and the second surface of the second semiconductor substrate is opposite to the first surface of the second semiconductor substrate; a wiring line; and a shield between the joining surface and the wiring line, wherein the shield includes a first via and a second via, an end of the first via is connected to the second surface of the first semiconductor substrate, and an end of the second via is connected to the second surface of the second semiconductor substrate.

2. The solid-state imaging element according to claim 1, further comprising: a third via that penetrates the first semiconductor substrate; and a fourth via that penetrates the second semiconductor substrate, wherein the wiring line is configured to connect the third via to the fourth via, and the first semiconductor circuit is connected to the second semiconductor circuit through the third via, the fourth via, and the wiring line.

3. The solid-state imaging element according to claim 2, wherein the wiring line is on the first semiconductor substrate and the second semiconductor substrate.

4. The solid-state imaging element according to claim 2, further comprising a support substrate joined to the second surface of the first semiconductor substrate and the second surface of the second semiconductor substrate, wherein each of the third via and the fourth via penetrates the

support substrate, and the wiring line is on the support substrate.

5. The solid-state imaging element according to claim 2, further comprising a wiring layer that includes the wiring line, wherein the wiring layer corresponds to a redistribution layer, and fabrication of the wiring layer is based on a plating technique.

6. The solid-state imaging element according to claim 2, further comprising a wiring layer that includes the wiring line, wherein fabrication of the wiring layer is based on a photolithography technique and a dry etching technique.

7. The solid-state imaging element according to claim 1, wherein the first semiconductor substrate is a logic substrate, the first semiconductor circuit is a logic circuit, the second semiconductor substrate is a memory substrate, and the second semiconductor circuit is including a memory circuit.

8. The solid-state imaging element according to claim 1, wherein the first semiconductor substrate is a first logic substrate, the first semiconductor circuit is a first logic circuit, the second semiconductor substrate is a second logic substrate, and the second semiconductor circuit is a second logic circuit.

9. The solid-state imaging element according to claim 1, wherein the first semiconductor substrate is a first memory substrate, the first semiconductor circuit is a first memory circuit, the second semiconductor substrate is a second memory substrate, and the second semiconductor circuit is a second memory circuit.

10. The solid-state imaging element according to claim 1, further comprising a first memory substrate that includes a first memory circuit, wherein the first memory substrate is joined to the pixel substrate, the first semiconductor circuit is connected to the first memory circuit, and the second semiconductor circuit is connected to the first memory circuit.

11. The solid-state imaging element according to claim 1, further comprising: a first wiring line configured to connect the first semiconductor circuit with the pixel circuit; and a second wiring line configured to connect the second semiconductor circuit with the pixel circuit, wherein the first wiring line is on the first semiconductor substrate, and the second wiring line is on the second semiconductor substrate.

12. The solid-state imaging element according to claim 10, wherein the first semiconductor substrate is a logic substrate, the first semiconductor circuit is a logic circuit, the second semiconductor substrate is a second memory substrate, and the second semiconductor circuit is second memory circuit.

13. The solid-state imaging element according to claim 4, further comprising: a first wiring line configured to connect the first semiconductor circuit with the pixel circuit; and a second wiring line configured to connect the second semiconductor circuit with the pixel circuit, wherein the first wiring line is on the first surface of the first semiconductor substrate, and the second wiring line is on the first surface of the second semiconductor substrate.
